

Problem

The voltage across the terminals of a circuit is

$$v(t) = [30 + 20 \cos(60\pi t + 45^\circ) + 10 \cos(120\pi t - 45^\circ)] \text{ V}$$

If the current entering the terminal at higher potential is

$$i(t) = 6 + 4 \cos(60\pi t + 10^\circ) - 2 \cos(120\pi t - 60^\circ) \text{ A}$$

find:

(a) the rms value of the voltage,

(b) the rms value of the current.

(c) the average power absorbed by the circuit.

Step-by-step solution

Step 1 of 4

Consider a circuit with the voltage and current in the form as,

The voltage across the terminals is,

$$v(t) = [30 + 20 \cos(60\pi t + 45^\circ) + 10 \cos(120\pi t - 45^\circ)] \text{ V}$$

The current entering the terminals is,

$$i(t) = [6 + 4 \cos(60\pi t + 10^\circ) - 2 \cos(120\pi t - 60^\circ)] \text{ A}$$

Step 2 of 4

a) The root mean square value of the voltage is,

$$\begin{aligned} V_{rms} &= \sqrt{a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} A_n^2} \\ &= \sqrt{(30)^2 + \frac{1}{2} [(20)^2 + (10)^2]} \\ &= \sqrt{900 + \frac{1}{2} [400 + 100]} \\ &= \sqrt{900 + \frac{1}{2} [500]} \\ &= \sqrt{900 + 250} \\ &= \sqrt{1150} \\ &= 33.91164 \text{ V} \end{aligned}$$

The root mean square value of the voltage is 33.91164 V.

Step 3 of 4

b) The root mean square value of the current is,

$$\begin{aligned}
 I_{rms} &= \sqrt{a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} A_n^2} \\
 &= \sqrt{(6)^2 + \frac{1}{2} [(4)^2 + (-2)^2]} \\
 &= \sqrt{36 + \frac{1}{2} [16 + 4]} \\
 &= \sqrt{36 + \frac{1}{2} [20]} \\
 &= \sqrt{36 + 10} \\
 &= \sqrt{46} \\
 &= 6.78232 \text{ A}
 \end{aligned}$$

The root mean square value of the current is 6.782329 A.

Step 4 of 4

c) The average power absorbed by the circuit is,

$$\begin{aligned}
 P &= V_{dc} I_{dc} + \frac{1}{2} \sum_{n=1}^{\infty} V_n I_n \cos(\theta_n - \phi_n) \\
 &= (30)(6) + \frac{1}{2} [(20)(4) \cos[-45^\circ - (-10^\circ)] + (10)(-2) \cos(45^\circ - 60^\circ)] \\
 &= 180 + \frac{1}{2} [80 \cos(-45^\circ + 10^\circ) - 20 \cos(45^\circ - 60^\circ)] \\
 &= 180 + \frac{1}{2} [80 \cos(-35^\circ) - 20 \cos(-15^\circ)] \\
 &= 180 + \frac{1}{2} [80 \cos(35^\circ) - 20 \cos(15^\circ)] \\
 &= 180 + \frac{1}{2} [80(0.8191520443) - 20(0.9659258263)] \\
 &= 180 + \frac{1}{2} [65.53216354 - 19.31851653] \\
 &= 180 + \frac{1}{2} [46.21364701] \\
 &= 180 + 23.10682351 \\
 &= 203.1068235 \text{ W}
 \end{aligned}$$

The average power absorbed by the circuit is 203.1068235 W.